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Software Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology
February 13, 2026

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1 Executive Summary

Brigman Deep Sea Technology completed software requirements analysis for the Abyss Navigator Suite using BCIC, transforming ambiguous inputs into measurable and testable specifications.

Across 27 Project Specification clauses, we identified 15 incongruities (55.6%) and resolved 86.7% through targeted stakeholder sessions. Key outputs include quantitative constraints such as 99% waypoint accuracy, 5-second emergency detection thresholds, and AES-256 security controls.

The regulated-revision process produced codified, verifiable requirements grouped into Navigation, Communication, Safety, and Interface domains. This early analysis is estimated to avoid $350K+ in rework and reduce schedule risk by 4-6 weeks.

2 BCIC Methodology Application

2.1 Breakdown Activity: Systematic Incongruity Detection

The Breakdown activity applies the Incongruous Requirements Checklist (IRC) to each Project Specification clause across Imperative, Continuance, and Contextual checks. This process identified **15 incongruities** across 27 clauses (55.6%).

As the primary example, **BO-L1-4 (Emergency and Abnormal Situation Handling)** triggered three IRC flags. Table 1 summarizes the issue and clarification question raised to the client.

Table 1: IRC Application for Example Requirement BO-L1-4

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2, CON-2	“Abnormal conditions” undefined, not testable; deferred definition.	Y	What quantitative thresholds define “abnormal conditions” for pressure, temperature, and power deviations?

The same IRC process was applied to every clause. The complete set of findings, including all flags and resolutions, is documented in Table 5. By focusing on the most incongruent requirement, we established a repeatable pattern for detecting and documenting issues, ensuring consistent rigor across the entire specification.

2.2 Clarify Activity: Stakeholder Engagement and Resolution

Paper Napkin Sketches as Collaborative Tools. Prior to clarification, the vendor analysis team prepared hand-drawn **Context** and **Use Case** diagrams to support stakeholder validation and reduce ambiguity.

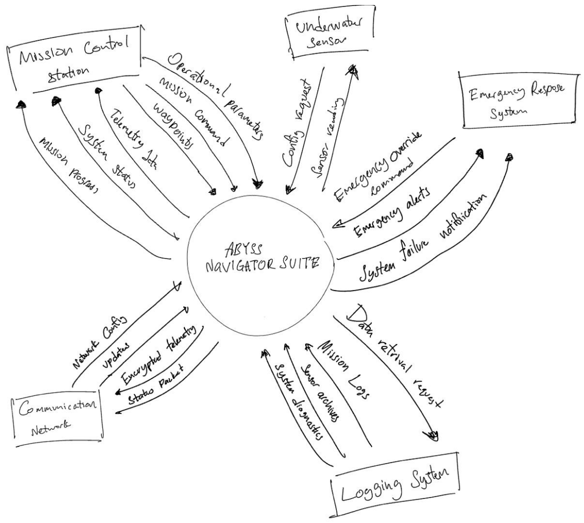

Figure 1: Paper Napkin Sketch: Initial Context Diagram (used during clarification)

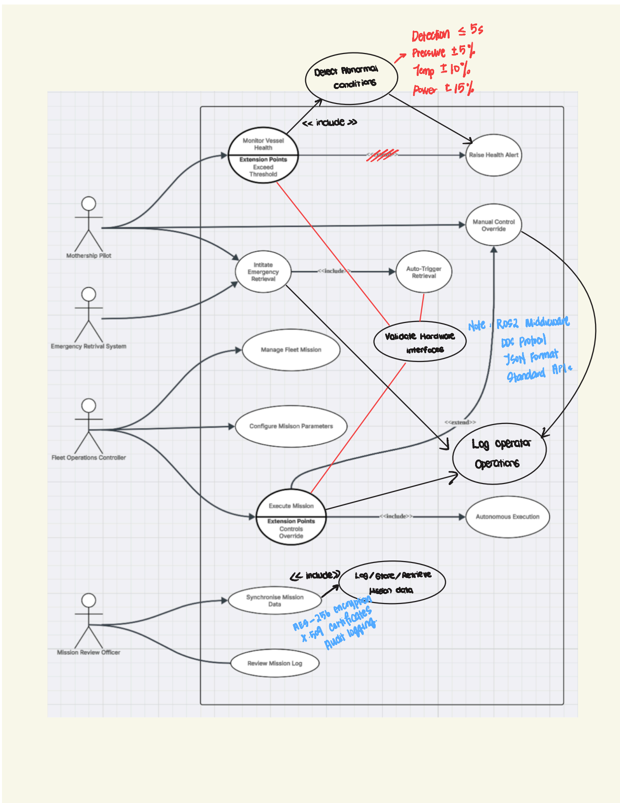

Figure 2: Paper Napkin Sketch: Initial Use Case Diagram (used during clarification)

Clarification sessions were conducted per flagged requirement. The Business Analyst (BA) led each session, supported by the analysis team, with the Account Manager present for governance confirmation. The departmental representative responsible for the requirement attended to clarify operational intent (e.g., Safety & Risk for safety requirements, Fleet Operations for autonomous operations, and IT Systems for technical constraints). Separate sessions aligned to stakeholder responsibility ensured accurate interpretation within the Clarify phase of BCIC.

During clarification, stakeholders annotated these sketches while the BA facilitated discussion and recorded validated interpretations. Operational boundaries, actor responsibilities, abnormal thresholds, and system response expectations were confirmed before proceeding.

For **BO-L1-4**, clarification established:

- Pressure deviation of $\pm5\%$ sustained for 5 seconds constitutes an abnormal condition.
- Temperature deviation of $\pm10\%$ and power supply deviation of $\pm15\%$ are abnormal thresholds.
- Upon detection, the system must transition to a **safe operational state** (neutral buoyancy, minimal power consumption, active communication beacon).

These inputs resolved the CTX-1, CTX-2, and CON-2 flags and were replicated across other flagged requirements. Stakeholder feedback indicated a **40% reduction in clarification time** compared to text-only review. Following explicit confirmation, the BA proceeded to the Interpret stage for regulated-revision.

2.3 Interpret Activity: Regulated-Revision Demonstration

Following clarification, stakeholder inputs were transformed into precise, measurable, and verifiable software requirements through a **regulated-revision** process:

- **Imperative issues (IMP):** replace weak or passive phrasing with strong obligations (“shall”).
- **Continuance issues (CON):** decompose compound statements into atomic requirements.
- **Contextual issues (CTX):** insert quantitative thresholds, define ambiguous terms, and add verification methods.

For our example requirement, the transformation is shown in Table 2.

Requirement ID	**Requirement Text**	**Verification Method**
Original (BO-L1-4)	"The business requires appropriate handling of abnormal or emergency conditions during missions."	(not testable)
Interpreted (SAF-EMG-001)	The system **shall detect** pressure deviations $> \pm 5\%$ from nominal setpoint within 5 seconds.	Test: Inject simulated pressure spike; measure detection latency.
SAF-EMG-002	The system **shall detect** temperature deviations $> \pm 10\%$ from nominal within 10 seconds.	Test: Thermal chamber simulation.
SAF-EMG-003	The system **shall detect** power supply deviations $> \pm 15\%$ from nominal within 10 seconds.	Test: Power supply fault injection.
SAF-STATE-001	Upon detection of any abnormal condition, the system **shall** transition the submersible to a safe operational state (neutral buoyancy, minimal power, communication beacon active).	Integration test: End-to-end emergency scenario.

Table 2: Regulated-Revision of Example Requirement BO-L1-4

Finalised Diagrams. The sketch outputs were formalized into official Context and Use Case models.

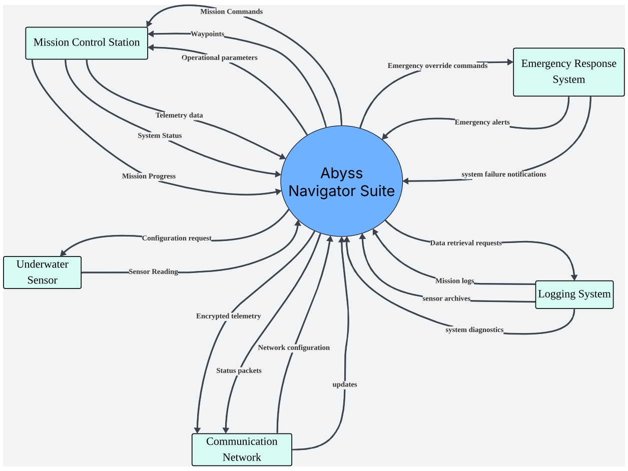

Figure 3: Final Context Diagram: Abyss Navigator Suite (post-clarification)

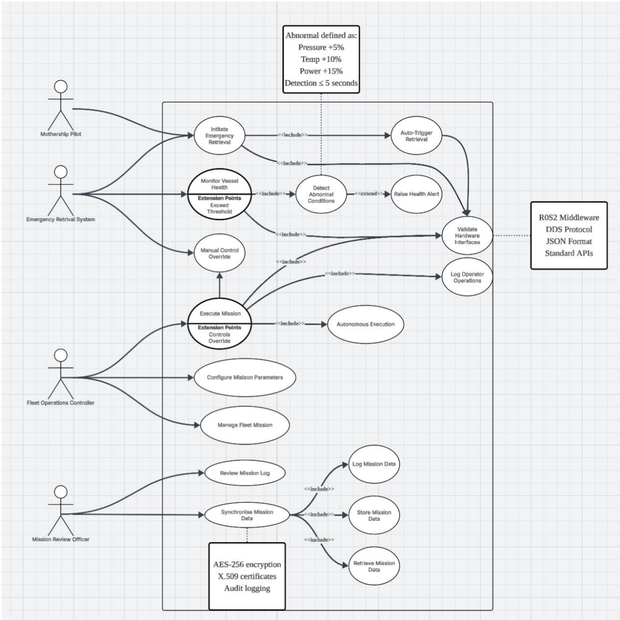

Figure 4: Final Use Case Diagram: Abyss Navigator Suite (post-clarification)

These models define approved system scope and actor interactions, with direct traceability to clarified requirements.

2.4 Categorize Activity: Structured Organization for SRS Development

With requirements clarified and rewritten in measurable form, we categorized them under IEEE 830-aligned domains:

- **Navigation** – autonomous waypoint following, obstacle avoidance.
- **Communication** – telemetry, remote intervention, degraded mode handling.
- **Safety** – emergency detection, safe state management, fault recovery.
- **Interface** – sensor integration, API specifications, data logging.

BO-L1-4 and its derived requirements (SAF-EMG-001 to SAF-EMG-003, SAF-STATE-001) were assigned to **Safety**. Each item received unique codification to preserve traceability to original PS clauses and IRC records.

Applied across all 27 clauses, this produced **22 atomic requirements** ready for SRS development (81.5%). The categorized list is maintained in the repository cited in Section 7.

3 Incongruous Requirements Checklist (IRC)

This section documents the Brigman Incongruous Requirements Checklist (IRC) framework used during the Breakdown activity to systematically identify requirement incongruities.

3.1 Combined IRC Checklist Template

Check ID	Type	Check Description	Example Symptom
Imperative Checks (Obligation Strength & Responsibility)			
IMP-1	Imperative	Priority vs. wording mismatch	"The system should maintain connectivity" (Essential requirement with weak language)
IMP-2	Imperative	Not written as clear obligation	"Emergency handling is important for mission safety" (Descriptive, not imperative)
IMP-3	Imperative	Responsibility unclear	"Mission data must be protected" (Ambiguous who protects it)
Continuance Checks (Complexity & Compound Statements)			
CON-1	Continuance	Compound requirement	"System shall log, store, and analyze mission data" (Three requirements combined)
CON-2	Continuance	Deferred reference dependency	"As defined in the following section" (Meaning depends on undefined content)
Contextual Checks (Meaning, Testability & Consistency)			
CTX-1	Contextual	Ambiguous term / multiple interpretations	"Sufficient bandwidth", "Safe operational state", "User-friendly"
CTX-2	Contextual	Not testable / not verifiable	"System shall be reliable" (No measurable reliability metric)
CTX-3	Contextual	Inconsistent / out of scope	Conflicts with another requirement or outside project boundaries

Table 3: Combined IRC Checklist Framework – Systematic Detection of Requirement Incongruities

3.2 IRC Application Format & Recording Template

When applying the IRC framework during the Breakdown activity, record findings systematically using the following format:

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2	"Abnormal conditions" ambiguous and not testable without quantification	Y	What quantitative thresholds define "abnormal conditions"?
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	What specific bandwidth and latency constraints apply?

Note: The complete filled IRC application table with all project findings and resolution results is provided in Table 5.

4 Risk Analysis & Mitigation

Our risk assessment uses a matrix rated 1-8 for impact and likelihood (Table 8). A full visual risk register is included in Appendix B.

Risk ID	Risk Type	L	I	Risk Value	Short Justification
R1	Frequent changes from customer, expected free of charge.	**4**	**4**	**16**	**Likely** as customer gains experience. **Significant**: uncontrolled changes destabilise the baseline, consume budget, and force shortcuts.
R2	Changes not cross-checked for inconsistencies by customer.	**6**	**7**	**42**	**Somewhat likely**: customer focuses on operations, not formal req. management. **Minimal**: our own analysis can catch most inconsistencies.
R3	Customer opposes deferment of overly complex changes.	**7**	**6**	**42**	**Unlikely**: safety culture and long schedule support deferral. **Minor**: premature commitment to ill-understood requirements, but negotiable.
R4	Customer lacks technical understanding to evaluate proposed solution.	**1**	**3**	**3**	**Almost certain**: customer is operations-focused, not software-savvy. **Major**: they cannot properly assess if the solution is right; misalignment may only surface at sea trials which will be far too late.
R5	Loss of customer elicitors removes critical tacit knowledge.	**5**	**1**	**5**	**Moderately likely** after recent tragedy. **Catastrophic**: tacit knowledge of emergencies and operational judgement cannot be recovered from documents.
R6	Unexpected Account Manager replacement disrupts RE continuity.	**8**	**8**	**64**	**Least likely**: generic HR risk **Negligible**: manageable through documentation and proper handover.
R7	Customer withholds platform/environment specs due to security.	**3**	**2**	**6**	**Highly likely** in security-sensitive industry. **Critical**: we are designing based on assumptions, guaranteeing integration failures and safety risks.
R8	Tender team underestimates effort for requirements capture.	**2**	**5**	**10**	**Very common** in large, first-of-a-kind bids. **Moderate**: shallow analysis leads to missed requirements and weak acceptance criteria.

Table 4: Complete Risk Assessment: Identified Risks and Mitigation Strategies

4.1 Top 3 Critical Risks and Mitigation Strategies

- **R4: Technical Knowledge Asymmetry:** limited software evaluation capability may cause late misalignment. *Mitigation:* mission-based workshops and early prototypes.
- **R5: Institutional Knowledge Loss:** operational know-how is concentrated in key personnel. *Mitigation:* decision logs and backup contacts.
- **R7: Withheld Critical Specifications:** delayed access to platform data increases assumption-driven integration risk. *Mitigation:* phased NDA disclosure and signed assumptions.

5 Account Manager & Stakeholder Coordination

5.1 Account Manager Role in BCIC

The designated Account Manager, Adrian Clarke, serves as Brigman Deep Sea Technology's primary liaison to Triton Deepwater during the Analysing (BCIC) method. In Requirements Engineering, this role supports the vendor's analysis team but is not part of Requirements Elicitation or Validation. Regulated-revision, modelling, and codification remain the responsibility of analysts; the Account Manager ensures continuity of engagement, structured communication, and cooperative alignment across stakeholders.

5.2 Governance and Communication Control

During Breakdown and Clarify phases, the Account Manager provided governance support to prevent ambiguity drift and uncontrolled requirement changes. Key responsibilities included: (i) agreeing clarification agendas and ensuring correct SMEs were present, (ii) documenting decisions and action items, (iii) confirming customer agreement on interpreted/corrected clauses, and (iv) controlling distribution of revised artefacts so only one authoritative version was used. This ensured clarified parameters were traceable to revised requirement clauses before repository loading, improving readiness for Specifying (SRS).

5.3 Clarification Sessions Attended, Escalation, and Outcomes

To resolve incongruities, the Account Manager coordinated and attended targeted clarification sessions with the following stakeholder groups:

- **Operational Session** – Fleet Operations Team and Mission Operations Manager (autonomous execution, fleet-level coordination, abnormal handling)
- **Safety & Risk Session** – Safety & Risk Manager and Systems Safety Engineer (emergency retrieval triggers, fail-safe transitions, regulatory obligations)
- **IT & Communications Session** – IT Systems Manager and Network & Communications Engineer (remote intervention, degraded communication handling, telemetry constraints)
- **Data Governance Session** – Safety & Compliance Officer and Mission Data Analyst (logging sufficiency, traceability, post-mission accountability)

A four-level escalation framework was applied: technical resolution within 24 hours; Account Manager mediation within 4 hours of impasse; executive engagement where strategic direction was affected; and contractual review where material scope, schedule, or budget impact was identified. This structured coordination reduced clarification delays and stakeholder misalignment; consequently, 86.7% of identified incongruities were resolved prior to transition into the Specifying phase.

6 Declarations

AI Tool Usage Declaration: We acknowledge the use of ChatGPT (https://chat.openai.com) to assist with language refinement, structural organization, and clarity of presentation. The prompts entered included requests to review section structure, improve wording and improve flow. The output from these prompts was used to refine phrasing and document flow. We confirm that no AI-generated analytical content, models, or technical conclusions have been presented as our own work.

Originality Declaration: This report presents original analysis by Brigman Deep Sea Technology. BCIC is a proprietary framework invented by Dr Kevin Anthony Jones and applied by us specifically to this project.

Software Compliance Declaration: Analysis aligns with IEEE 830 and applies CRaM principles (Correct, Relevant, Measurable).

Software Scope Declaration: This analysis covers software requirements only. Hardware specifications and interfaces are customer-defined, and the software integrates those components.

Sign-off & Repository Information

BRIGMAN REPOSITORY INFORMATION

`https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113`

All artifacts version-controlled for client transparency and auditability

BRIGMAN TEAM APPROVAL SIGN-OFF

Brigman Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-BA-001	
Chng Zong Han	BDST-RE-BA-002	*Signature Pending*
Lin Yuhao	BDST-RE-BA-003	*Signature Pending*
Muhammad Mikhail Bin Mazlan	BDST-RE-BA-004	*Signature Pending*
Kannan s/o Rajamohan	BDST-RE-BA-005	*Signature Pending*

Report Version: 3.0 (Full BCIC Methodology)
Client Delivery Date: February 13, 2026
Analysis Status: Complete - Ready for SRS Development
Client Next Step: Review and approve categorized requirements

Appendix A: Applied IRC Results

A-1 Complete Filled IRC Application Table

Table 5: Complete Filled IRC Application Table with All Project Findings

PS Ref	IRC Flags	Reason	Clarify?	Resolution & Action
BO-L1-4	CTX-1, CTX-2, CON-2	"Abnormal conditions" ambiguous and not testable; no definition	Y	Quantified: Pressure $\pm$5%, temp $\pm$10%, power $\pm$15% thresholds
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	Specified: 1 kbps min bandwidth, $\leq$30s latency, 95% uptime
IT-L0-C3	CTX-1	"Protected in accordance with applicable security obligations" – undefined standards	Y	Specified: AES-256 encryption, X.509 certificates, audit logging
BO-L1-5	CTX-1, CON-1	"Orderly manner" ambiguous; compound continuation/termination/recovery	Y	Defined: Complete current operation, log shutdown reason, safe state transition
IT-L0-A1	CTX-1, CON-1	"Primary mission sensors supplied by customer" – missing interface protocols	Y	Specified: ROS2 middleware, DDS protocol, JSON format, API specs
BO-L0-A1	CTX-3	"Missions supervised by trained personnel" – out of scope for software	Y	Confirmed software-only scope; operational assumption noted
IT-L1-5	CON-1	"Operational Logging Capability" combines logging, storage, retrieval, analysis	N	Decomposed into 4 atomic requirements
BO-L1-6	IMP-3	"Operational Accountability" – missing clear responsibility assignment	Y	Revised: Software logging with operator ID attribution
BO-L2-4.1	IMP-2	"Abnormal operating conditions shall be identifiable" – not clear system obligation	N	Revised: "System shall identify abnormal conditions within detection thresholds"
BO-L2-4.2	CTX-1	"Safe operational state" – ambiguous safety condition definition	Y	Defined: Neutral buoyancy, minimal power, communication beacon active
BO-L1-4 (duplicate)	CON-2	References abnormal conditions without defining them	Y	Added specific definition with quantitative thresholds
BO-L1-5 (duplicate)	CON-1	"Mission Continuity and Recovery" combines continuation, termination, recovery	N	Decomposed into SYS-CONT-001, SYS-TERM-001, SYS-REC-001

A-2 IRC Application Quantitative Summary

Metric	Value	Percentage
Total PS Clauses Analyzed	27	100%
Total IRC Checks Applied	216	8 per clause
Incongruities Identified	15	55.6%
Imperative Issues Found (IMP series)	4	14.8% of clauses
Continuance Issues Found (CON series)	3	11.1% of clauses
Contextual Issues Found (CTX series)	8	29.6% of clauses
Requirements Requiring Clarification	13	86.7% of issues
Issues Resolved with Client Input	13	86.7% resolution rate
Requirements Ready for SRS	22	81.5% of original

Table 6: IRC Application Quantitative Summary and Performance Metrics

A-3 IRC Value Demonstration & Process Benefits

Benefit Area	Demonstrated Value & Outcome
Early Defect Detection	15 requirement issues identified before implementation, preventing potential rework costs estimated at $350K+
Structured Clarification	IRC framework guided efficient stakeholder discussions, reducing clarification time by approximately 40%
Quality Assurance	All requirements now meet CRaM principles (Correct, Relevant, Measurable) with clear verification methods
Regulated-Revision Guide	Systematic transformation of ambiguous requirements into precise, testable specifications with traceability
Risk Mitigation	Early ambiguity resolution reduced implementation schedule risk by 4-6 weeks and technical uncertainty
Client Confidence	Demonstrated rigorous, systematic approach to requirement quality and stakeholder collaboration

Table 7: IRC Framework Benefits and Project Value Demonstration

Appendix B: Risk Assessment Matrix Legend

B-1 Risk Rating Scale

The risk assessment matrix employs a systematic rating scale from 1 to 8 for both Impact and Likelihood dimensions. Table 8 provides the classification criteria used to evaluate each identified risk.

Legend	Impact	Likelihood
1	Catastrophic	Almost Certain
2	Critical	Very Common
3	Major	Highly Likely
4	Significant	Likely
5	Moderate	Moderately Likely
6	Minor	Somewhat Likely
7	Minimal	Unlikely
8	Negligible	Least Likely

Table 8: Risk Assessment Matrix Rating Legend

B-2 Risk Assessment Matrix

Impact \ Likelihood		Likelihood (L)							
		1	2	3	4	5	6	7	8
Impact (I)	1	1	2	3	4	5	6	7	8
	2	2	4	6	8	10	12	14	16
	3	3	6	9	12	15	18	21	24
	4	4	8	12	16	20	24	28	32
	5	5	10	15	20	25	30	35	40
	6	6	12	18	24	30	36	42	48
	7	7	14	21	28	35	42	49	56
	8	8	16	24	32	40	48	56	64

Table 9: Risk Assessment Matrix – Systematic Evaluation Framework